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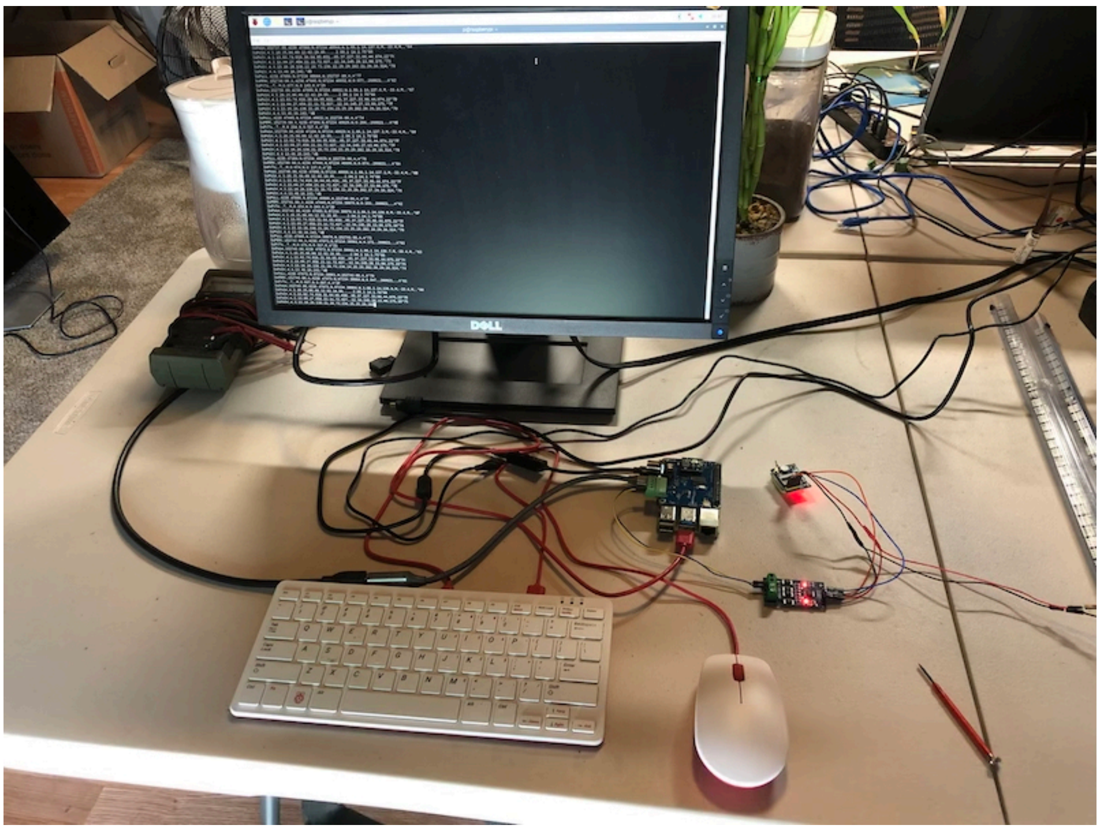
Testing NMEA 0183 For The PICAN-M - NMEA 0183 & NMEA 2000 HAT For Raspberry Pi

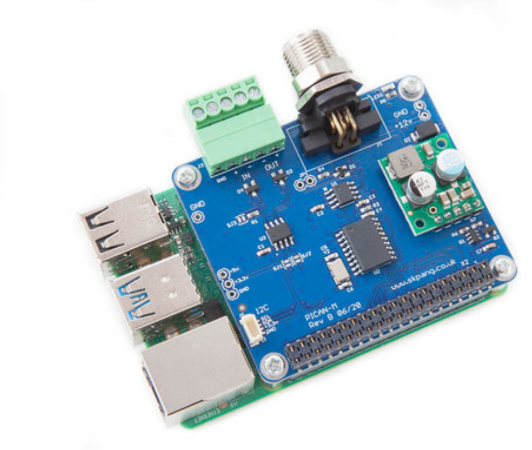
Posted by Wilfried Voss on Aug 26th 2021



PICAN-M - NMEA 0183 & NMEA 2000 HAT For Raspberry Pi

Our PICAN-M (M = Marine) is a Raspberry Pi HAT with NMEA 0183 and NMEA 2000 connection. The NMEA 0183 (RS422) port is accessible via a 5-way screw terminal. The NMEA 2000 port is accessible via a Micro-C connector.

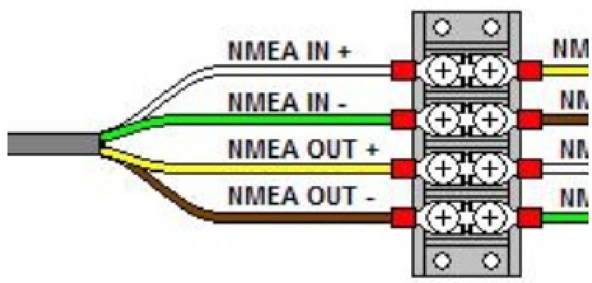
The board comes with a 3A SMPS (Switch Mode Power Supply), allowing to power the Raspberry Pi plus HAT from an onboard power source (12 VDC).



While most users engage in NMEA 2000 ap
older NMEA 0183 devices, and some of t
hardware and receiving data through the PICA
details of my setup and how to test the NMEA
basics to help understand the technology.

Copperhill Technologies is a member of
Association (NMEA).

A Brief Introduction To NMEA



Note: The above connection diagram, copied from the NMEA.org website, shows a wrong connectic
EIA-422 (RS 422). For more information see below paragraph *Connecting a NME 0183 Device*.

NMEA 0183 is a combined electrical and data specification for communication between marine electro
anemometer, gyrocompass, autopilot, GPS receivers and many other types of instruments. It has be
the National Marine Electronics Association. It replaces the earlier NMEA 0180 and NMEA 0182 stand
it is slowly being phased out in favor of the newer NMEA 2000 standard, though NMEA0183 remains th

The electrical standard that is used is EIA-422, although most hardware with NMEA-0183 outputs are
port. Although the standard calls for isolated inputs and outputs, there are various series of har
requirement.

The NMEA 0183 standard uses a simple ASCII, serial communications protocol that defines how data a
one "talker" to multiple "listeners" at a time. Through the use of intermediate expanders, a talker can
with a nearly unlimited number of listeners, and using multiplexers, multiple sensors can talk to a single

At the application layer, the standard also defines the contents of each sentence (message) type, so th
accurately.

The NMEA standard is proprietary and sells for at least US$2000 (except for members of the NMEA
much of it has been reverse-engineered from public sources.

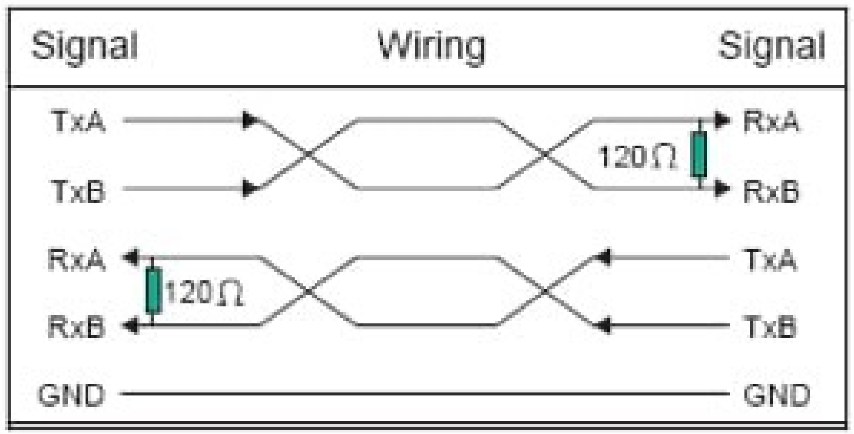
More Resources

- NMEA 0183 (Wikipedia)...
- NMEA 0183 Interface Standard (NMEA.org)...
- Everything you need to know about NMEA 0183 (PDF)...
- NMEA 0183 Installation And Operating Guidelines (PDF)...

- The NMEA 0183 Protocol (PDF)...

A Brief Introduction To RS-422

RS-422 is a standard for serial data transfer similar to RS-232, but it uses a differential voltage, i.e., the 232 uses a reference to ground). RS-422 uses twisted pair (difference pair) to represent the logic represents so-called balanced transmission as it does not refer to ground, thus providing a noise-pro both lines, which are differentiated out, allowing to carry data at considerably longer distances and transmits data to up to 1200 meters (~3600 feet). The maximum transfer rate is 10 Mbits/s.

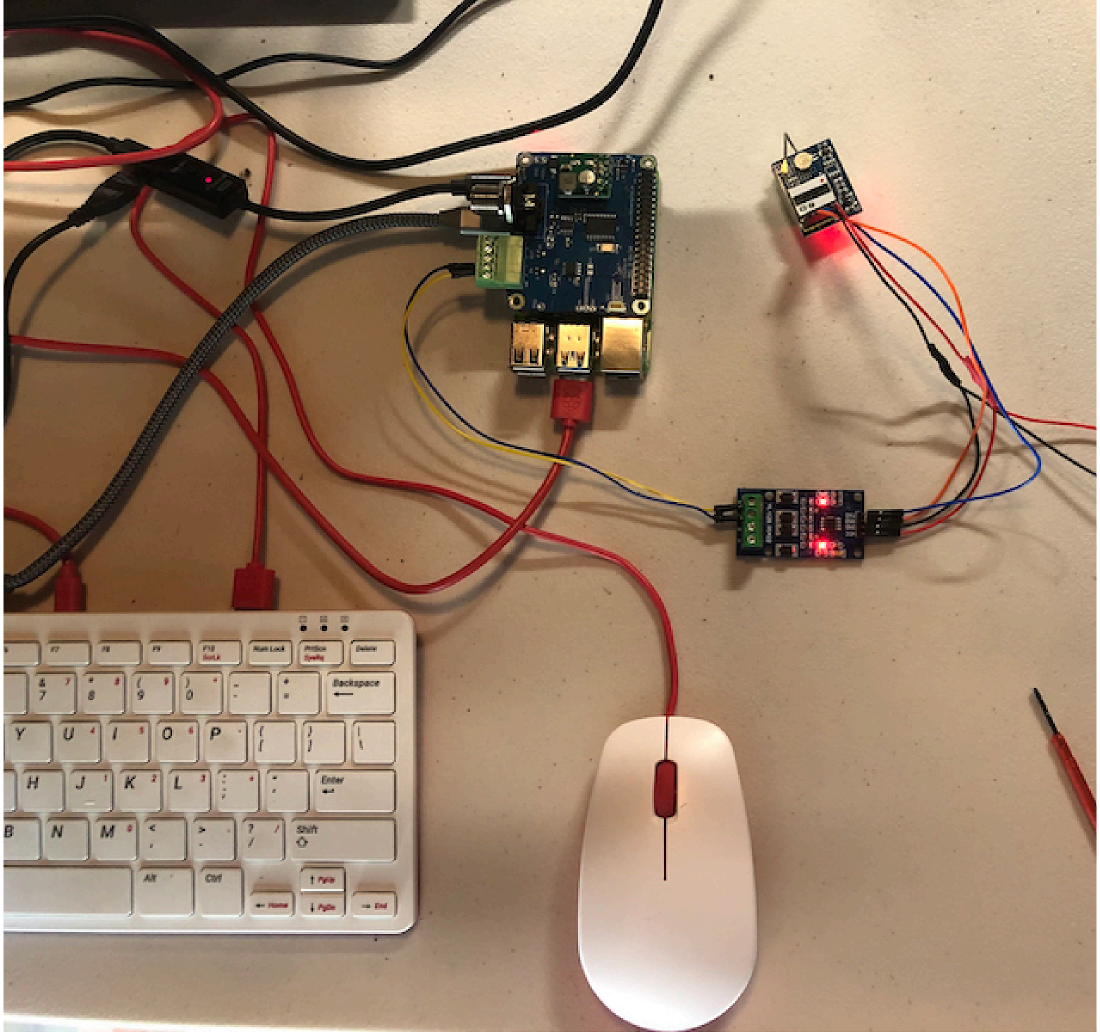


A 120 Ohm resistor acts as a terminal resistor that removes reflections during transmission over long twisted pair lines are 4 VDC and between transmission lines is 12 VDC. Therefore, RS-422 can be cc by simply connecting the negative wire of the twisted pairs to the ground.

Typically, the differential signals are referred to as RX+, RX- lines for receive and TX+,TX− lines for tra sometimes referred to as RX-A, RX-B, and TX-A, TX-B, or as the inverted and non-inverted signals. U does not specify the nomenclature of the particular signals, only the signal levels. This means that e choose their signal names and pinouts. Due to this, care must be taken when connecting two devi matched correctly.

The Test Setup

Looking at the above wiring diagrams for NMEA 0183 and RS-422, you may notice a discrepancy wire/signal description. The confusion worsens with RS-422 devices that use A, B, Z, and Y to descrit post's topic is about NMEA 0183, and connecting NMEA 0183 devices is (or should be) straightforwar as RS-422 since the hardware on the PICAN-M port is just that. Also, I was not in the mood to buy an instead, I used our UART GPS Module with Real-Time Clock in combination with our RS422 to TTL image.

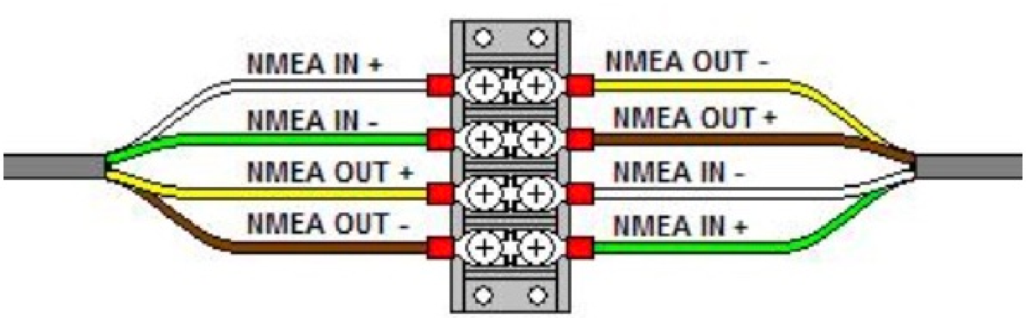


Connecting a NMEA 0183 Device

As I mentioned previously, connecting an NMEA 0183 should be straightforward, but it does happen th
requirements. I encountered the same problem as some of my customers: The data output as displa
garbage, but that was solved by swapping the + and - signals.

**As indicated in the above NMEA 0183 wiring diagram, IN+ connects to OUT+ and IN- connects t
it does not comply with the EIA-422 (RS 422) Standard.** It may be that the official Standard contair
ready to spend $1000 for it.

Looking at the above RS-422 wiring diagram, TxA (+) connects to RxB (-), and TxB (-) connects to R
results in:



Note that I changed the polarity (+/-) on the right-hand side.

Preparing the Raspberry Pi

Before you can test the data traffic, you need to prepare the Raspberry Pi to set up the serial interface. Manual (pages 8 and 9) for the setup, explicitly adding the line **enable_uart=1** and the **raspi-config**

After connecting your NEMA0183 device to the green connector of the PiCAN-M board, type the follow

stty -F /dev/ttyS0 raw 4800 cs8 clocal

cat /dev/ttyS0

You should now see the data traffic on the screen.

My Specific Setup

As I wrote before, I am using our GPS module with TTL signal output in combination with an RS42 RS422-compatible. However, this setup came with its own challenges, especially with the RS422 modu

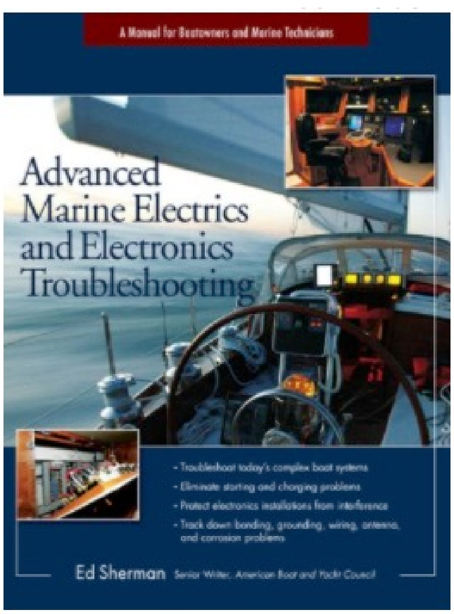
First, there was the connection from the GPS device to the RS422 board that did not follow the stan and RX-TX.

- GPS-RX connects to RS422-RSD (most probably a design mishap)
- GPS-TX connects to RS422-TXD

Secondly, the RS422 signals (A, B, Z, Y) follow neither the official nor the NMEA 0183 standard, and w Y, it was impossible to find a cross-reference on the Internet. Nevertheless, I figured it out:

- A = IN+
- B = IN-
- Z = OUT+
- Y = OUT-

Last, but not least, I had to use a different baudrate (9600), which I applied to the stty command. The (to 4800 baud, but for the proof of concept that was not necessary.



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